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Executive Report: Taxi Fare Data Preparation

Project: Automatidata | Client: New York City Taxi & Limousine Commission (TLC)
PACE Phase: Analyze (Data Preparation and Structuring)

Business Context

- The NYC Taxi & Limousine Commission (TLC) is the agency responsible for licensing over 200,000 taxi and for-hire vehicles. They oversee nearly 1 million trips daily, generating a massive and complex dataset
- Operational Volume: The agency oversees approximately one million trips daily, generating a massive and complex dataset.
- Strategic Challenge: Despite the vast amount of information, the TLC faces the challenge of transforming historical data into tools that improve user experience and fare transparency.

Project Overview

- The objective of this project is to transform TLC's historical data into a data-driven solution that allows users to estimate their trip fares before boarding. Currently, we have completed the initial phase of data loading and structuring to ensure the information is processable and reliable.

Business Questions

- What is the relationship between trip distance and the final fare?
- Are there significant fare variations based on the time of day or day of the week?
- How do geographical zones impact the total cost?
- What percentage of the fare is represented by additional charges like tolls and tips?

PACE Methodology

We have adopted the PACE framework to ensure ethical and professional development:

- Plan: Defined project goals and identified key stakeholders (Juliana Soto, Titus Nelson, DeShawn Washington, and Luana Rodriguez).
- Analyze: Focused on data cleaning, outlier detection, and structuring the dataset into a "Tidy Data" format using Python..
- Construct (Upcoming): Development of a Multiple Linear Regression model to predict taxi fares based on trip distance and duration.
- Execute (Upcoming): Presentation of final insights and deployment of the fare estimation tool for the TLC.

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IMPACT

By identifying structural anomalies and organizing the dataset early, we ensure a high-quality 'Tidy Data' foundation. This reduces future errors in the taxi fare prediction modeling phase and provides a clear roadmap for a Deep Dive Exploratory Data Analysis (EDA), ultimately leading to more precise and transparent fare estimations for the TLC."

Key Insights

After inspecting the 18 variables in the dataset, we identified that trip_distance and total_amount are the most critical factors for the initial regression model. Preliminary inspection revealed significant outliers that require immediate cleaning to prevent model bias.

Anomalous Data Reference

- Irregular Fare Patterns: We identified "unusual values" where trips with zero or near-zero distance have disproportionately high charges (e.g., up to $450.00).
- Skewed Distributions: Similar to other transport datasets, fare amounts and distances are heavily right-skewed, necessitating the use of median statistics to ensure robust and reliable conclusions.
- Payment Type Correlation: A preliminary link was found between payment methods and recorded totals, suggesting that tip reporting varies significantly between credit card and cash transactions..
- Data Integrity: Identifying these outliers is essential to ensure that the final Multiple Linear Regression model is not influenced by erroneous data points.

trip_distance (Miles)	fare_amount ($)	Analysis / Observation
2.60	999.99	Extreme Outlier / Potential Error
0.00	450.00	Zero Distance / High Fare
0.00	200.00	Inconsistent Data
5.20	22.50	Standard Relationship (Normal)

"...suggesting that identifying and treating extreme fare-to-distance anomalies is the primary requirement for a reliable and ethical estimation tool."

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Project Governance & Roles

To ensure project success, we have defined specific roles and responsibilities aligned with the PACE framework, guaranteeing clear communication between technical teams and institutional stakeholders.

Role / Rol	Assignee / Asignado	Responsibility / Responsabilidad
Ethical Mentor / Mentor Ético	**Juliana Barrientos**	Strategic oversight and ethical narrative. / Supervisión estratégica y narrativa ética.
Data Analyst / Analista	**DeShawn/ Marcelo Dominguez**	Data exploration and modeling. / Exploración de datos y modelado.
Stakeholders (TLC) / Institucionales	**Titus Anneke**	Goal validation and decision making. / Validación de objetivos y toma de decisiones.

Next Steps

Deep-Dive Exploratory Data Analysis (EDA):

- Conduct a comprehensive analysis of the validated dataset to identify core correlations between trip_distance, time_of_day, and total_fare.
- Generate visual distributions to isolate and understand the impact of the previously identified fare anomalies

Advanced Feature Engineering:

- Prioritize variables with a direct impact on trip costs.
- Evaluate "Data Boosters" such as duration_estimates, vendor_analysis, and peak_hour_indicators to enrich the model's predictive power before training.

Model Selection & Evaluation:

- Evaluate XGBoost and Multiple Linear Regression as primary candidates for the "Construct" phase.
- Compare performance metrics (MAE, RMSE, R^2) to select the most robust algorithm for automated fare estimation.

Documentación y Visualización:

- Update the project repository on GitHub with the cleaned "Tidy Data" structure.
- Design specialized dashboards to communicate insights to both technical and non-technical stakeholders, including Juliana Soto and DeShawn Washington.